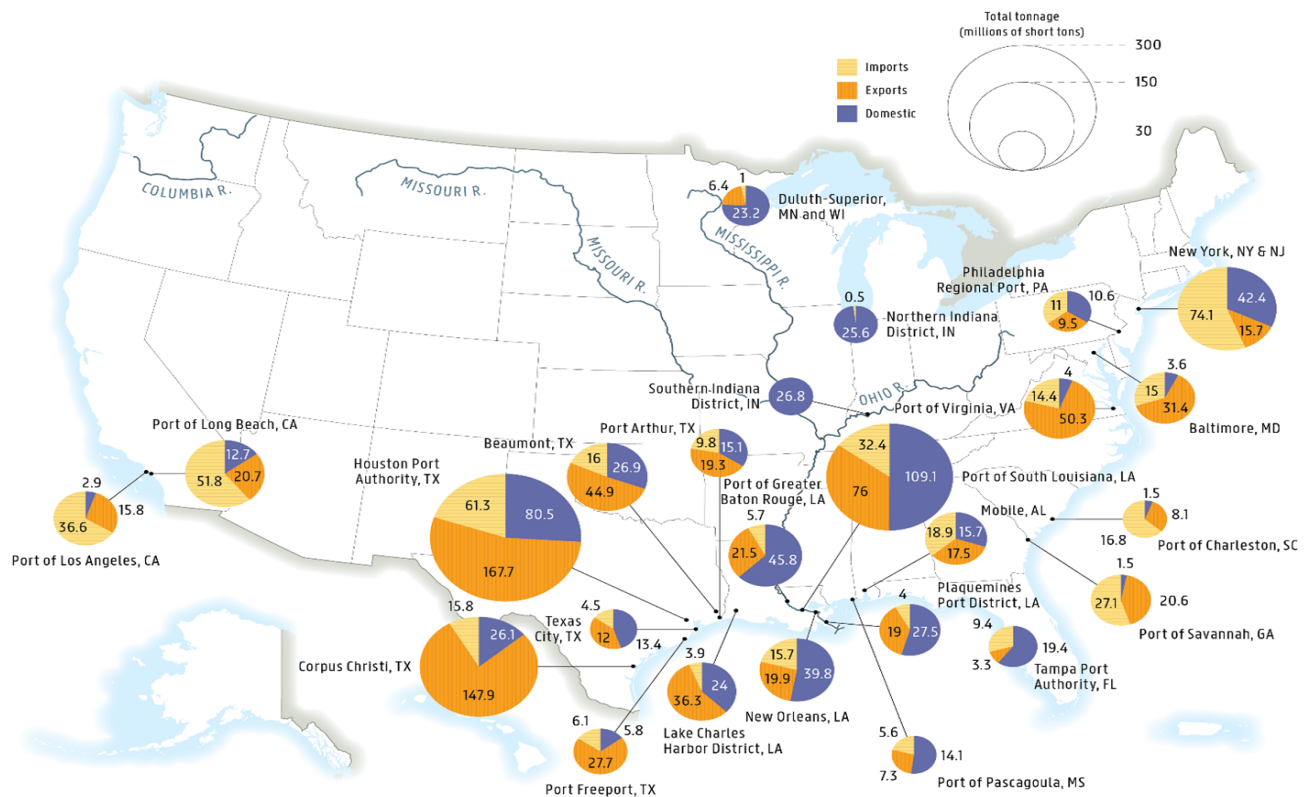




ADA ACCESSIBILITY ANALYSIS REPORT

1. Background

This visualization is a U.S. map containing dozens of pie charts representing cargo tonnage at major ports. Each pie displays the share of Imports (yellow), Exports (orange), and Domestic (blue). Circle size indicates relative total tonnage. Labels identify each port, and numeric values appear inside pie slices.

2. Text Contrast

- Most state labels, river labels, and port names have adequate contrast against a light background.
- Some very small numeric labels inside pie slices approach borderline contrast due to their size and placement over strong colors.

3. Graphic Contrast

- The three categorical colors (yellow, orange, blue) maintain acceptable visual contrast.
- Some small wedges (e.g., very thin Import or Export slices) visually blend at small display sizes.
- No anti-aliasing contrast failures severe enough to impact interpretability were detected.

4. Colorblind Simulation

- Chart remains broadly interpretable under simulated Deuteranopia, Protanopia, and Tritanopia conditions.
- Yellow vs. orange becomes less distinguishable for red-green colorblind users, but the domestic blue remains clearly distinct.
- Heavy reliance on color means the legend must remain close to the graphic for accessibility.

5. OCR / Text Legibility

- Many numeric labels are < 9 pt visual equivalent in screen rendering, which does not meet WCAG low vision guidance for body text.
- Port labels remain readable at normal zoom but are too small for accessibility without magnification.

6. DPI / Sharpness

- Image renders cleanly with high clarity (vector-like).
- No blur or pixelation issues affecting comprehension were observed.

7. Recommended Alt Text

“Map of the United States showing major port locations, each represented by a pie chart indicating the distribution of imports (yellow), exports (orange), and domestic tonnage (blue). The size of each circle represents total cargo volume in millions of short tons. Large volume ports include Houston, New Orleans, and New York/New Jersey. Labels provide tonnage values for each slice.”